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Bias in Emotion Recognition: How Accent Shapes AI's Perception of Speech

Introduction and Data

1.1. Motivation and Context

The increasing deployment of Artificial Intelligence (AI) in high-stakes professional settings, such as virtual interview assistance and candidate screenings, necessitates a rigorous examination of algorithmic fairness and accuracy. Many of these services incorporate emotion recognition modules that claim to objectively assess a candidate's emotional state based on vocal tone. However, the use of such technology raises profound ethical concerns regarding algorithmic bias, particularly when assessing communication styles tied to demographic characteristics like accent, ethnicity, and age. The prevalence of services like Sanas and Krisp highlights a societal preference for certain vocal norms, compelling us to question the reliability of classification systems applied to diverse voices.

This project focuses specifically on the susceptibility of emotion recognition models to biases associated with a speaker's accent. Prior research has established that models in domains like facial recognition perpetuate widespread racial biases due to non-representative training data (Scheuerman et al., 2021). We hypothesize that a similar issue exists within speech-based AI. While accent is not race, it serves as a powerful linguistic marker conveying geographic or ethnic origin, placing it in a similar sphere of potential bias. Critically, the human tendency to perceive

speech through the lens of accent bias provides a strong theoretical foundation for this study. As demonstrated by Jiang et al. (2019), the presence of an unfamiliar or “outgroup” accent can significantly alter how human listeners interpret speech. In the context of AI, this bias is likely due to data imbalance, where models, trained predominantly on voices with a majority accent (e.g., General American English), struggle to accurately classify emotion in underrepresented accents, leading to systematic misclassification and potentially unfair outcomes.

This investigation became particularly interesting when our final model training run demonstrated a testing categorical accuracy of approximately 59.17% (with a training categorical accuracy of 67.10%), significantly higher than previous runs, suggesting the model is consistently learning emotion-related patterns.

However, this increased performance revealed a critical instability related to bias detection:

- The more sensitive Chi-squared test for specific accents (English, Spanish, Mandarin) sometimes showed statistical significance, suggesting the model's predictions and accent are not entirely independent
- This result was unstable, shifting between significant and insignificant depending on the training run
- When conducting post-hoc tests with a corrected, stricter alpha value (0.01667) to account for multiple comparisons, none of the pairwise comparisons were statistically significant

This fluctuation highlights a key finding: the extent of model overfitting directly impacts classification outcomes for different accents. The model's sensitivity to slight changes in training

weight updates suggests that its predictive certainty is unstable when dealing with diverse accents.

These questions, while addressing unfamiliar accents, valence bias, and general demographic bias, drove our decision to combine and standardize datasets, forming a rigorous methodology designed to test algorithmic fairness against the real-world complexity of human diversity.

1.2. Research Questions and Hypotheses

This study seeks to answer the following questions:

- How does a single, trained emotion-classifying model respond to speech exhibiting unfamiliar accents?
- Is there a bias in the classification of basic emotional valences (e.g., negative vs. positive) across different accents when analyzed by an emotion-recognition model?

We hypothesize that models trained primarily on current emotional speech datasets will exhibit a significant bias, attributing a higher rate of negative emotions (such as anger and sadness) to voices with non-American accents compared to voices with a predominant American accent. We believe this discrepancy will be driven by the model's unfamiliarity with the vocal characteristics of diverse accents.

1.3. Data Description and Preparation

To address these questions, we employ two key audio datasets:

- Crema-D (Crowd-sourced Emotional Multimodal Actors Dataset): Provides 7,442 sound bites from 91 actors (aged 20–74) expressing six emotional states. This will be the primary source for training our emotion classification models.
- Speech Accent Archive: Contains 2,140 English audio samples from English speakers across 177 countries and 214 native languages. This serves as the novel test set for introducing diverse accents.

Data Preparation, Variable Transformations, and Model Inclusion

The most significant initial transformation is Feature Extraction: converting all raw audio files into a sequence of Mel-Frequency Cepstral Coefficients (MFCCs). MFCCs (typically 13 coefficients per audio frame) are the industry standard for emotional speech analysis, effectively representing the spectral envelope of the sound. All files are standardized to a uniform sampling rate and trimmed of silence to ensure feature consistency. This MFCC Sequence is the numerical input variable for our training model.

Metadata requires significant aggregation for statistical testing. Target Variable Standardization consolidates the six Crema-D emotions into three simplified Emotional Valence categories: Negative (Anger, Disgust, Fear, Sadness), Positive (Happiness), and Neutral. Separately, the raw geographic metadata from the Speech Accent Archive is tidied by mapping specific countries into a manageable set of Accent Groups (e.g., North American, South Asian, Slavic, Western European). This Accent Group variable served as the Explanatory Variable for the chi-square test, which we used to identify post-model bias. Samples with poor audio quality or missing critical demographic data are excluded from the final analysis set.

Methods

To evaluate the level of accent bias in the Crema-D dataset, we decided to create a neural network that predicts the emotions of recorded human speech. In order to accomplish this, we decided to rely on audio feature sequences that were successively fed into an LSTM. An LSTM (long short-term memory) is a type of recurrent neural network that keeps track of previous model outputs as it learns (Hager & DiPietro, 2020). By keeping track of its previous states and feeding those states back into the model, LSTMs are able to track patterns that are displayed over the course of successive model inputs (Hager & DiPietro, 2020). In audio, sequential patterns take the form of rhythm and intonation—factors which we expect to relate strongly to verbally displayed emotions. Since we plan to extract our audio data using spreads of numerical representations that span the course of the audio clip, an LSTM will allow us to identify these emotion-relevant variables.

In order to generate numerical representations of our audio files, we will utilize Mel-Frequency Cepstrum Coefficient features (or MFCCs). MFCCs are time-based power representations of sound timbre (Sidhu et al., 2025). By utilizing a mel-scale transformation, a scale that is more susceptible to low-frequency changes than higher frequency changes, MFCCs are able to mathematically describe sound in ways that mirror human sound perception. Additionally, MFCC's ability to describe sound timbre, a significant characteristic of human speech, makes it incredibly effective when attempting to train deep learning classifiers on human speech patterns (Sidhu et al., 2025).

In order to extract the MFCC data from each audio file in the Crema-D dataset, we utilized the Python-based librosa audio-analysis package. Since the librosa library's default values are already tuned for human speech analysis, we ended up preserving most of the default

MFCC-loading variables. The only variation from this was the value of MFCC features (for variable `n_mfcc`), which we set at 40 MFCC. We did this as we wanted to obtain a higher level of detail for each continuous audio clip, and thus, required a greater number of MFCC values.

As MFCC features were extracted from each audio file, each feature array was appended to a list that would eventually be used to fit the LSTM model. MFCC features in the list were then padded in order to ensure that each series of MFCCs was the same length (preventing audio clip length from disrupting the model's input size). This was done by utilizing Keras' preprocessing tool to add "0" values—values that are invisible to the model during training—to the beginning of each MFCC sequence. By using these tools, we were able to extract evenly-lengthened audio features from each dataset for both training and testing.

To create validation data for our LSTM model, we also had to extract emotion labels from the Crema-D dataset. This was completed by extracting the emotion labels from the path name of each Crema-D audio file. To allow the LSTM model to interact with the emotion-based validation data, we used TensorFlow and Keras to create categorical embeddings for each emotion. These embeddings would serve as the y-variable in our model.

Our LSTM model was created using the Keras and Tensorflow APIs. The architecture of the model is as follows: the model accepts an input whose shape matches the number of MFCCs defined when loading the audio file features using the librosa library (`n_mfcc`). It contains 2 LSTM layers (each with 250 nodes), an initial dense layer (with an l2 regularizer and a relu activation) with 100 nodes, a dropout layer with a value of 0.3, a second dense layer with 80 nodes (with an l2 regularizer and a relu activation), and a softmax-activated output layer that outputs (a one-hot encoding that describes) one of six possible emotions embeddings.

After creating the LSTM model, we compiled it with an Adam optimizer that was set to a learning rate of 0.001. We also established categorical cross-entropy and categorical accuracy as the loss and metric values, respectively. After compiling, the model was fit using 40 epochs and a batch size of 30. During training, input examples were shuffled in order to minimize overfitting caused by repeated audio clips taken from the same subject. We also implemented early stopping callbacks in order to allow the model to stop training on its own in cases of overfitting. Finally, a graph that displayed accuracy and loss values across all training epochs was plotted in order to visually confirm the extent of overfitting.

Ultimately, this model was used to generate emotion predictions on each of the Speech Accent Archive accent audio files. These predictions were then grouped together to form three graphs. The first two graphs were used to display the emotion counts for the ten most represented languages in the Speech Accent Archive (the first graph displayed the individual emotion counts for the top five languages, the second graph displayed the next five). Since the accent archive contained 200 languages total (with some receiving significantly fewer examples than others), we reasoned that collecting individual data for only the ten most represented languages would prevent accents with fewer recordings from skewing results. As a result, all other languages that were not represented in the ten most represented languages were grouped into a third and final graph. To account for differences in accent representation, we modified our three, count-based graphs to display relative frequencies (in terms of percentages of the accent's total representation) for each accent. This allowed us to more easily compare the LSTM's classifications of emotions between accents with vastly different sample sizes.

Finally, a statistical test was utilized in order to examine differences in emotion categorization between three target languages from the Speech Accent Archive. The first of these

languages, and the most represented language in the Speech Accent Archive, was English. Since the Crema-D dataset appeared to be dominated by American-accent speakers, examining the classification results for English speakers in the Speech Accent Archive provided us with information on how the model classified accents that are most similar to the training data. In addition to this, we examined Speech Accent Archive samples from two languages connected to underrepresented minorities in American society: Mandarin and Spanish. Mandarin was chosen in order to fill the role of an accent that may be prevalent (even if only a little) in the accents of the Crema-D dataset. Since the Crema-D dataset utilized a small sample of (unspecified) Asian speakers, we chose to use the most well-represented Asian accent in the Speech Accent Archive (Mandarin) to test whether unlabeled Crema-D accents influenced model training. Similarly, Spanish was chosen to represent the most common foreign accent in the Speech Accent Archive that was not represented in the Crema-D speaker demographics.

After identifying target languages for our analysis, a three-way, chi-squared test of homogeneity was run in order to identify significant differences between the emotional classifications provided by our Crema-D-trained model. Additional post-hoc testing was conducted as necessary in order to examine language-to-language differences.

Results

1. Demographic Imbalance in the CREMA-D Dataset

We first examined whether demographic variables (age, sex, and race) were evenly distributed across the six emotion categories in the CREMA-D dataset. These analyses were conducted using a test split.

1.1 Age Distribution

Across all emotions, the average speaker age falls within a range of approximately 35-37 years, indicating that the CREMA-D dataset is dominated by middle-aged speakers.

	Age
Emotion	
ANG	36.323887
DIS	35.494340
FEA	36.264925
HAP	35.745763
NEU	36.723502
SAD	36.597656

There is no significant age variation between the emotion classes. This suggests that age-related analyses might not produce useful results, being that most emotions are represented by speakers of nearly identical ages. This raises concerns about whether a model trained on this dataset is capable of generalizing to speakers of other age groups.

1.2 Gender Distribution

Sex	Female	Male
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Emotion		
ANG	119	128
DIS	115	150
FEA	133	135
HAP	123	113
NEU	99	118
SAD	123	133

The representation of Gender is not equal, and the imbalance varies by emotion. For example, neutral (NEU) and disgust (DIS) feature significantly more male speakers, while fear (HAP) is slightly female-dominant. These uneven patterns may lead the model to implicitly learn correlations between certain emotions and specific genders, potentially embedding gender bias.

1.3 Race Distribution

Race	African American	Asian	Caucasian	Unknown
Emotion				
ANG	56	16	171	4
DIS	70	29	165	1
FEA	58	23	185	2

CREMA-D is heavily dominated by Caucasian speakers, and the rest of the racial groups are vastly underrepresented. For each emotion category, Caucasian speakers make up approximately 65-75%, African American speakers account for about 20-25%, and Asian speakers only represent 5-10%. This imbalance makes it likely that the model will become better at gauging emotion for some races and generalizing poorly for underrepresented groups, leading to algorithmic racial bias.

2. LSTM Emotion Training Performance

To train the LSTM model, we extracted MFCC features from CREMA-D WAV files. Training accuracy improved steadily, reaching about 67.10%, and validation accuracy peaked at around 59.17%.

Epoch	Train Accuracy	Testing Accuracy	Train Loss	Testing Loss
40	0.6710	0.5917	0.9430	1.1761

The model output reveals a gap between training and validation accuracy, which suggests the existence of overfitting. This leads us to believe that although the model might learn the training distribution, it still struggles to generalize. If we recall the demographic imbalance mentioned

earlier, we can see how it might be attributed to the reliance on demographic details rather than purely emotional cues, especially if certain racial groups dominate specific emotional categories. That being said, the final testing data accuracy of 59.17% is significantly higher than any emotion-categorization accuracy one would expect from random chance. Since there are six emotions to classify, an unsuccessful, and randomly-guessing, model would have an accuracy closer to 16.66%. Since various training attempts of the model each ended up around 60%, we are much more likely to assume that our model is consistently learning emotion-related patterns.

3. LSTM Emotion Classification Performance

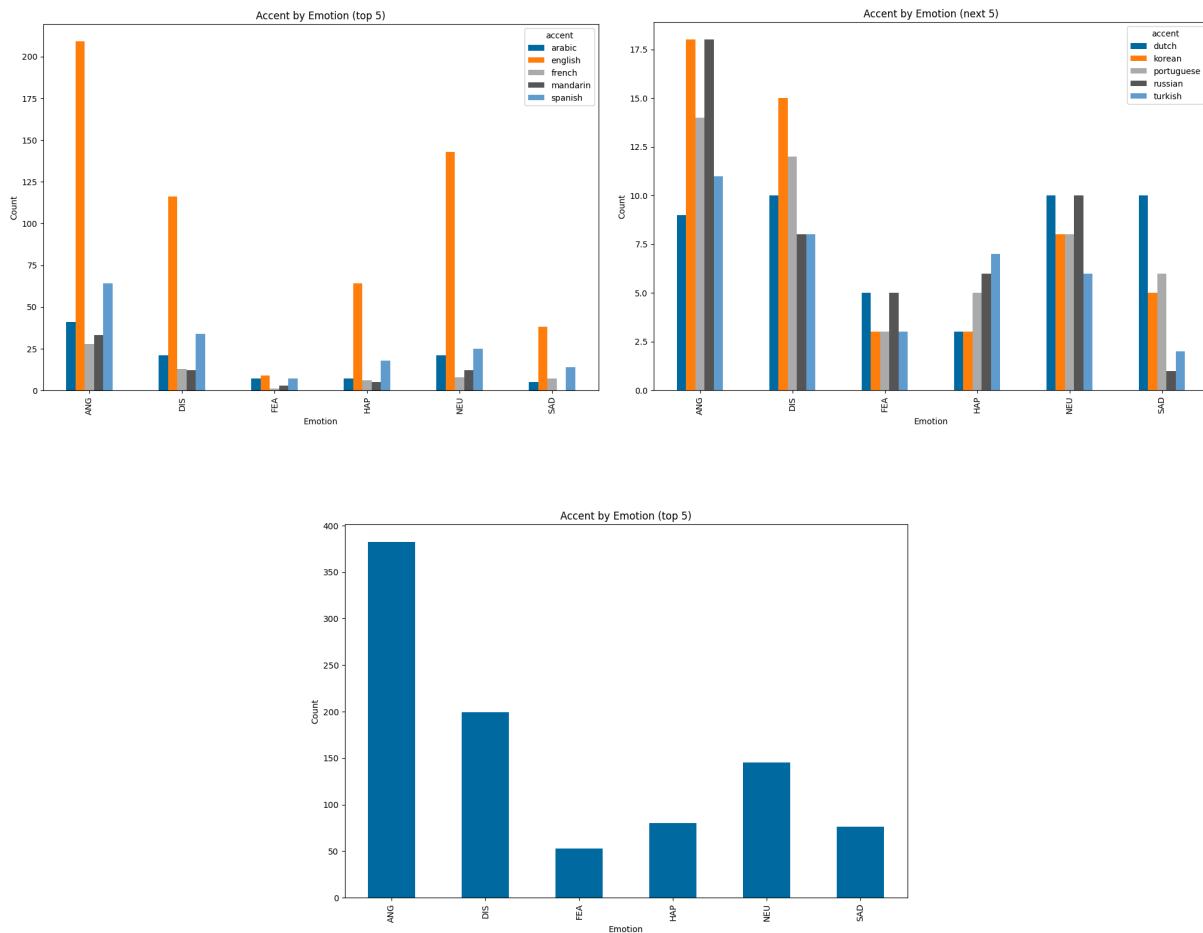


Figure 1: Graphs showing the counts of classified emotions in the top ten most represented languages in the Speech Accent Archive (top) and the counts of all other languages (bottom).

After training our model, we were able to apply it to the Speech Accent Archive dataset and create emotion categorizations. The total categorization counts can be found above in Figure 1. While most of these graphs seemed to follow similar patterns, the disproportionately high quantity of native English-speaking samples caused us to create new graphs that displayed the percentage of each emotion for each language.

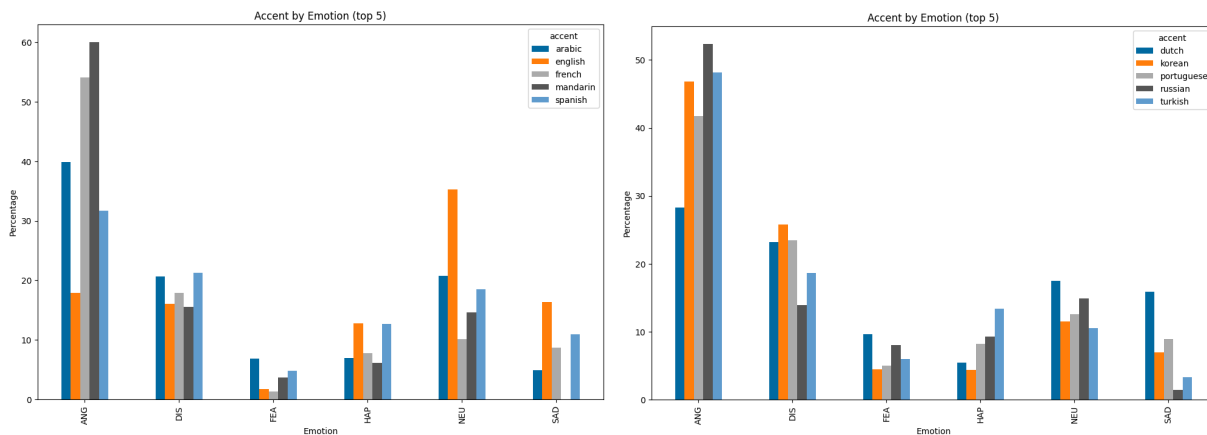


Figure 2: Graphs showing the relative frequencies of classified emotions in the top ten most represented languages in the Speech Accent Archive. Nearly all of the languages skew towards anger (ANG) as the primarily identified emotion, except for English, which seems to skew towards neutral (NEU)

After adjusting our graphs to account for sample sizes, we were able to notice a couple of key patterns that emerged between languages. For most languages, there seemed to be an emotion classification bias in favor of anger (ANG). Nearly all of the top languages followed this rule, including the general trends exhibited by the non-top-ten languages (see the bottom graph of Figure 1). That being said, English notably diverged from this pattern, appearing to be classified as predominantly neutral instead of angry.

In order to see whether this divergence was a product of improper model training, we

decided to retrain our emotion classifier and reclassify our audio files several times. While certain emotional frequencies would change in between trials—with English receiving varying anger and neutral scores—a noticeably lower relative anger percentage and a noticeably higher neutral percentage were always observed in the English sample when compared to other languages.

4. Accent-Based Evaluation

A chi-square test for homogeneity was conducted to examine if the distribution of emotional categorizations differed across English, Spanish, and Mandarin speakers in the Speech Accent Archive. The results showed a significant difference between the emotion distributions of the three languages, $X^2(10, N = 806) = 20.6120, p = 0.0240$.

	ANG	DIS	FEA	HAP	NEU	SAD
english	209	116	9	64	143	38
mandarin	33	12	3	5	12	0
spanish	64	34	7	18	25	14

Three additional chi-square post hoc tests were conducted to identify significant differences between the three languages. While using an alpha value that was corrected for three statistical tests (0.01667), no significant correlation was detected between English and Spanish ($X^2(5, N = 741) = 10.5166, p = 0.0619$), Spanish and Mandarin ($X^2(5, N = 227) = 7.9459, p = 0.1592$), or English and Mandarin ($X^2(5, N = 644) = 12.1858, p = 0.0323$).

Discussion

The overarching goal of this project was to investigate whether an emotion-recognition model trained on a widely used emotional speech dataset displayed systematic biases when exposed to voices with varying accents. Although the accent-based evaluation is still in progress, the results we have obtained thus far have already revealed several important insights about model behavior, dataset characteristics, and broader implications for fairness and transparency in speech-based AI.

Our analyses of the CREMA-D dataset reveal significant demographic imbalance across race, gender, and age. The data is made up primarily of Caucasian speakers, with most being in their mid-30s. Gender is unevenly represented as well, with specific emotions portrayed by one gender more than the other. This suggests that the model is trained on demographically limited data, and likely causes it to rely on correlations between emotion and demographics, and not necessarily on acoustic patterns relevant to emotion.

The accuracy results of the LSTM reinforce this concern. While the training accuracy reached roughly 74.7%, our final model iterations achieved a training accuracy of about 67% and a testing accuracy of about 59%. These findings suggest overfitting to the training population, although not nearly as severe as earlier attempts. Then, when the model operates based on these demographic correlations as opposed to emotion itself, the results become unreliable. The model's instability can then be identified even before accent variation is introduced, as per our first research question. Since the training data is demographically narrow, the model connects its understanding of emotion to the vocal patterns of the majority group. This strongly implies that when we eventually input speech from actors with unfamiliar accents, the model is likely to misinterpret the speech due to its limited exposure to vocal diversity.

The second research question asks whether bias exists in the classification of basic emotional valence across accents. Our chi-squared analysis produced mixed results. The overall chi-square test across all accents did in fact show statistical significance (p equaling about 0.024), suggesting that model predictions and accent are not entirely independent. However, when we conducted post-hoc tests with a corrected alpha level (0.0167), none of the pairwise comparisons were statistically significant. This suggests that while the model's predictions vary significantly across accents, the differences between specific accent pairs may not be strong enough to be detected individually. Additionally, the slightly different chi-squared values suggest that if bias does exist, it's likely subtle.

Our third research question details whether emotion classification differs based on the actor's age. While the CREMA-D dataset contains speakers from ages ranging between 20 and 74, the distribution is so tightly clustered around the mid-30s that meaningful age comparison becomes nearly impossible. This severely limits our ability to detect any age-related bias. As a result, the model likely generalizes poorly with younger and older speakers, this time due to a lack of exposure rather than a learned bias.

To an extent, the discrepancies we have identified both within the Crema-D dataset and our resulting findings indicate a need for well-documented, transparent data. Despite advertising itself as a diverse dataset with a demographically diverse set of speakers, our data reveals biases in the dataset towards certain demographics. Ultimately, it is the culmination of these slight biases that requires data scientists to use multiple datasets in order to attain a large set of diverse data to train models. Ultimately, the acceptance of biased data and the attempts of data scientists to straighten results indicate a preference for model work over data work. According to Scheuerman et al. (2021), this tendency has become one of many growing trends in dataset

curation and data science. As also indicated by Scheuerman et al. (2021), however, this preference is dangerous as data is just as important to model outcomes as the architecture models train on and the scientists who implement the data. As a result, we echo Scheuerman et al.'s claim and argue that datasets deserve more transparency and attention.

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